

E-COMMERCE RETENTION ANALYTICS

RFM Segmentation & Churn Prediction

Capstone Project Report

Prepared by Aromal

Project Overview

Why this project, and what it delivers

1

Objective

Segment e-commerce customers by purchase behavior and predict which are at risk of churning, so retention efforts can be targeted where they matter most.

2

Dataset

DataCo Smart Supply Chain dataset (Kaggle) — 180,519 orders across 53 columns, aggregated to 20,641 unique customers.

3

Approach

RFM (Recency, Frequency, Monetary) feature engineering, K-Means segmentation, and a 3-model churn classification comparison.

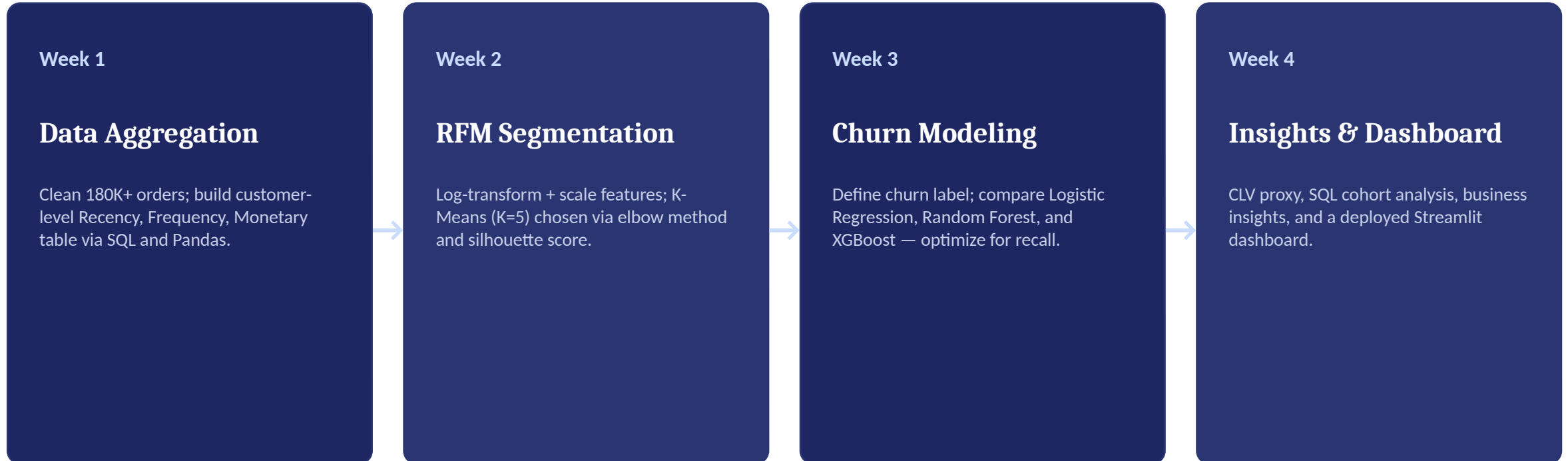
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Deliverable

A live, interactive Streamlit dashboard plus a prioritized action list of high-value customers at risk of churning.

Methodology

A four-week, four-stage pipeline



Dataset at a Glance

180,519

Orders analyzed

20,641

Unique customers

53

Original columns

3 yrs

Transaction history span

RFM Feature Definitions

- **Recency:** Days since a customer's most recent order
- **Frequency:** Count of distinct orders placed by the customer
- **Monetary:** Total sales value generated by the customer

Data Cleaning

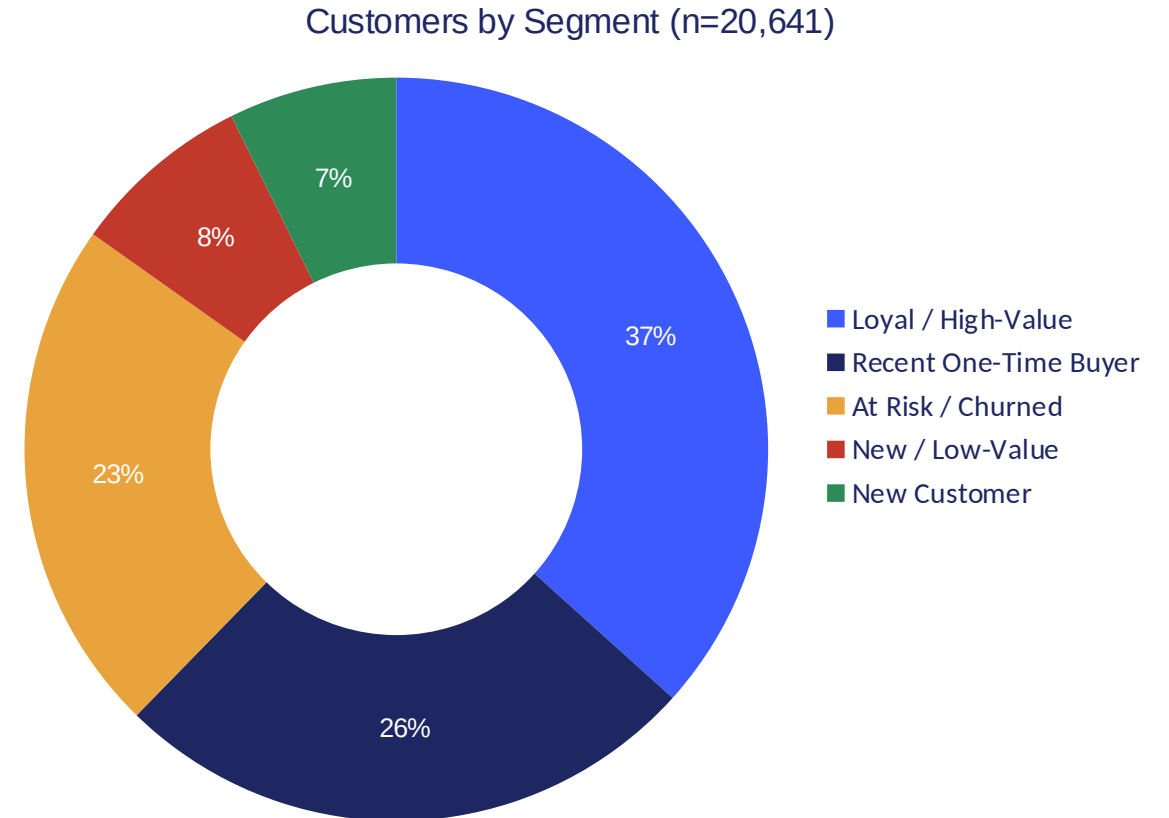
- Dropped fully-null and low-value columns (Product Description, Order Zipcode)
- Removed PII-style fields (Email, Password) as a precaution
- Dropped rows with missing Customer Lname / Zipcode (11 rows)
- Parsed order dates; handled Latin-1 encoding in source CSV

Customer Segmentation

K-Means clustering on log-transformed, standardized RFM features

How K = 5 was chosen

- Applied log1p transform to Recency, Frequency, Monetary to correct right-skew
- Standardized features (mean 0, std 1) so no single variable dominates distance
- Elbow method showed inertia flattening around K=4-5
- Silhouette score confirmed K=5 (0.458) as the best realistic choice



Segment Profiles

SQL cohort aggregation — average metrics per segment

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary	Avg CLV	Avg Churn Risk
Loyal / High-Value	7,568	254.1	5.9	\$3,573.79	\$22,685.04	40.5%
At Risk / Churned	4,648	450.2	2.7	\$1,554.33	\$4,571.35	76.0%
Recent One-Time Buyer	5,292	73.0	1.0	\$438.22	\$439.94	3.5%
New / Low-Value	1,623	77.8	1.0	\$37.84	\$37.84	2.0%
New Customer	1,510	11.0	1.0	\$86.02	\$86.02	2.3%

Key read: At Risk / Churned carries the second-highest average CLV (\$4,571) despite being only 22.5% of the base — meaningful revenue is quietly slipping away. Even the top Loyal / High-Value segment shows 40.5% average churn risk, so high past value alone doesn't guarantee retention.

Churn Modeling Approach

Defining the target, and avoiding data leakage

Churn Definition

Churned = 1 if a customer's Recency exceeds 307 days (the 75th percentile).

Result: a ~75% / 25% active/churned split, closely matching the size of the At Risk / Churned segment found independently via K-Means — a useful cross-check.

Feature Set

Used: Frequency, Monetary, Customer Tenure

Excluded: Recency — since it directly defines the churn label, including it would leak the answer into the model.

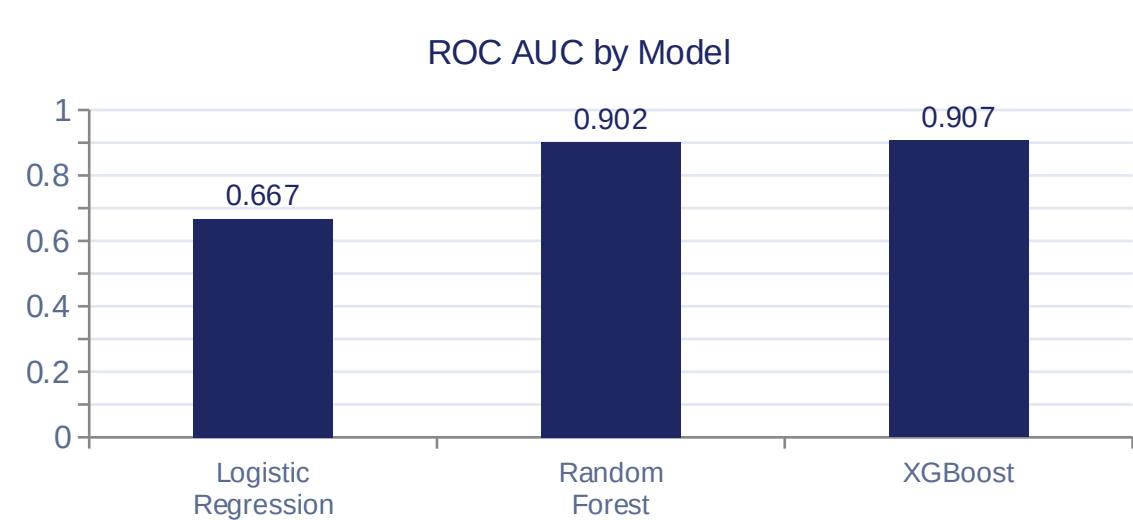
Why Recall Matters Here

For a retention program, missing an actual churner (false negative) costs more than a false alarm (false positive) — you simply skip a healthy customer in an outreach campaign. Model selection was optimized for Recall on the churned class, per the project brief.

Model Comparison

Three churn classifiers, evaluated on the same held-out test set

Model	Features	Precision (Churn)	Recall (Churn)	F1 (Churn)	ROC AUC
Logistic Regression	Freq, Monetary	0.34	0.56	0.42	0.667
Random Forest	+ Tenure	0.62	0.80	0.70	0.902
XGBoost (final)	+ Tenure	0.60	0.91	0.73	0.907

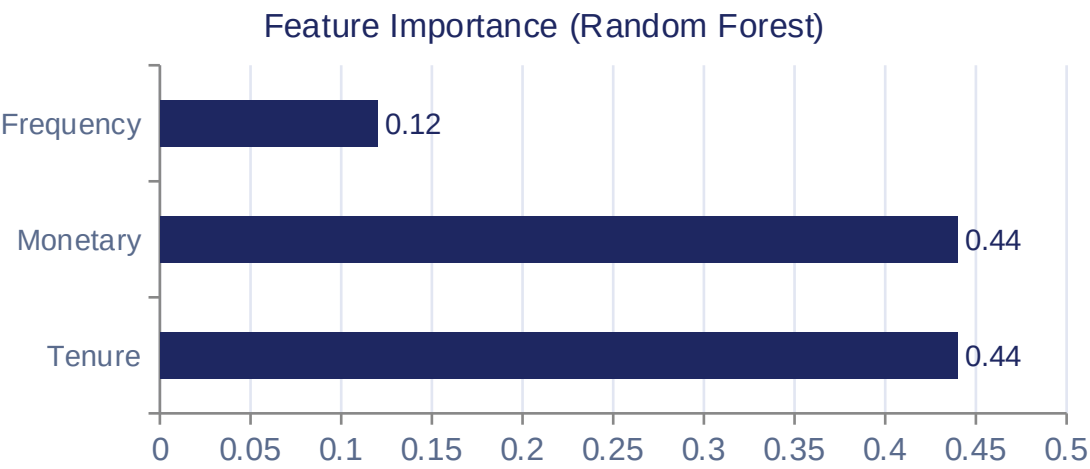


Why XGBoost Won

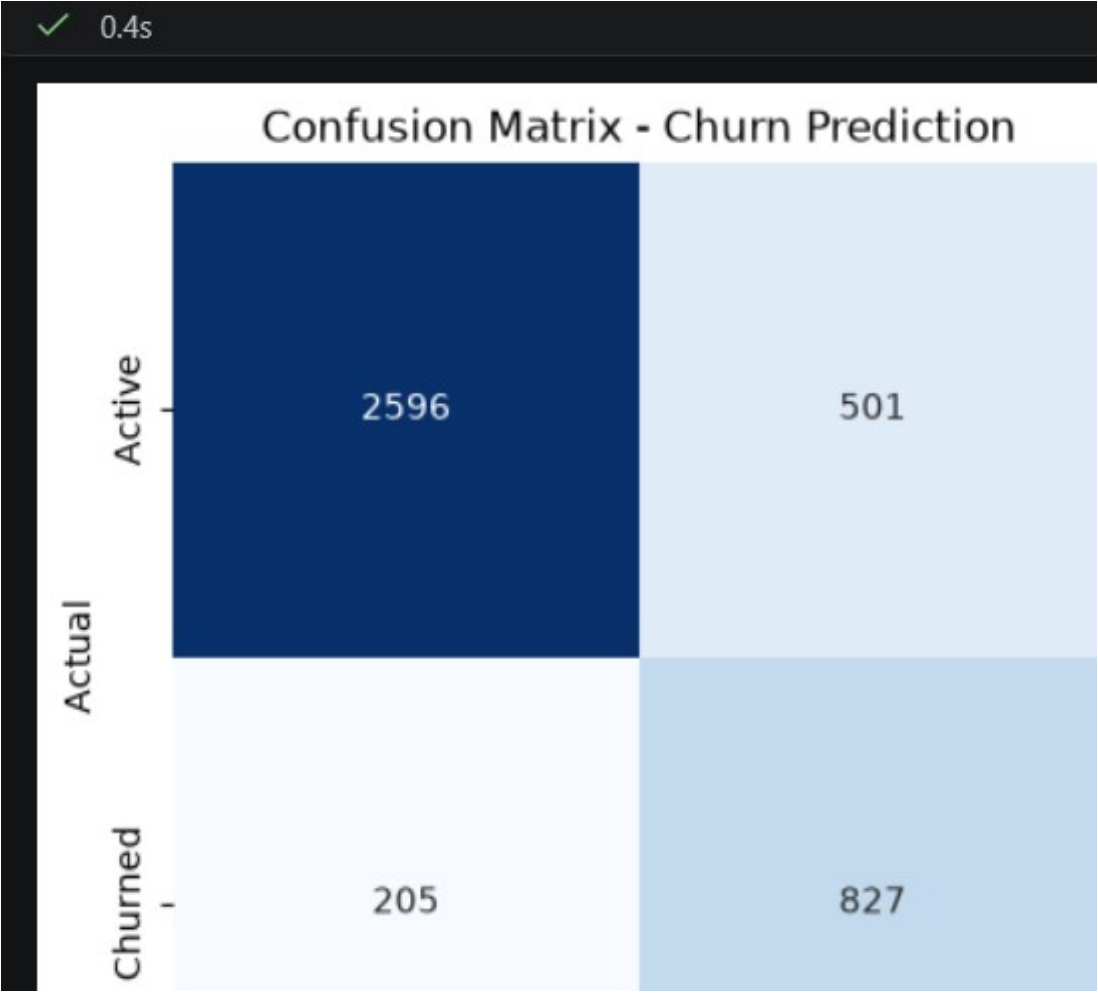
- Recall on churned customers jumped from 56% to 91% between the baseline and final model
- Best ROC AUC (0.907) among all three models tested
- Trade-off accepted: more false positives, in exchange for catching far more true churners

Final Model: What Drives Churn Risk

XGBoost feature importance and test-set confusion matrix



Tenure and Monetary contribute almost equally (44% each) to churn prediction — far outweighing raw order Frequency (12%). How long someone has been a customer, and how much they've spent, matter more than how often they buy.



Business Insights

1 of 2 — what the data reveals about retention risk

1

Revenue at risk, not just customer count

At Risk/Churned customers are 22.5% of the base but carry \$4,571 avg CLV — the 2nd-highest of any segment. This is real revenue slipping away, not low-value drift.

2

Even top customers aren't safe

Loyal/High-Value customers (top CLV at \$22,685) still show 40.5% average churn risk. Past value doesn't guarantee future loyalty — recency matters for everyone.

3

Tenure and spend predict churn, not order count

Tenure and Monetary each contribute ~44% feature importance, versus 12% for Frequency. How long and how much someone has bought matters more than how often.

4

The final model catches 91% of churners

XGBoost's high recall means the retention program will miss very few at-risk customers — the right trade-off when losing a customer costs more than one unnecessary email.

Business Insights

2 of 2 — opportunities alongside the risks

5

New customers aren't a churn concern — yet

New Customer and New/Low-Value segments show near-zero churn risk simply because they haven't had time to go quiet. They need onboarding-focused engagement, not churn intervention.

6

A large conversion opportunity exists

Recent One-Time Buyers (5,292 customers, 25.6% of the base) are recent but single-purchase. Nudging a second order could shift them into the Loyal segment.

7

CLV highlights a priority 'whale' group

Regardless of behavioral segment, customers in the top 10% of CLV proxy warrant dedicated retention attention — losing one has outsized revenue impact.

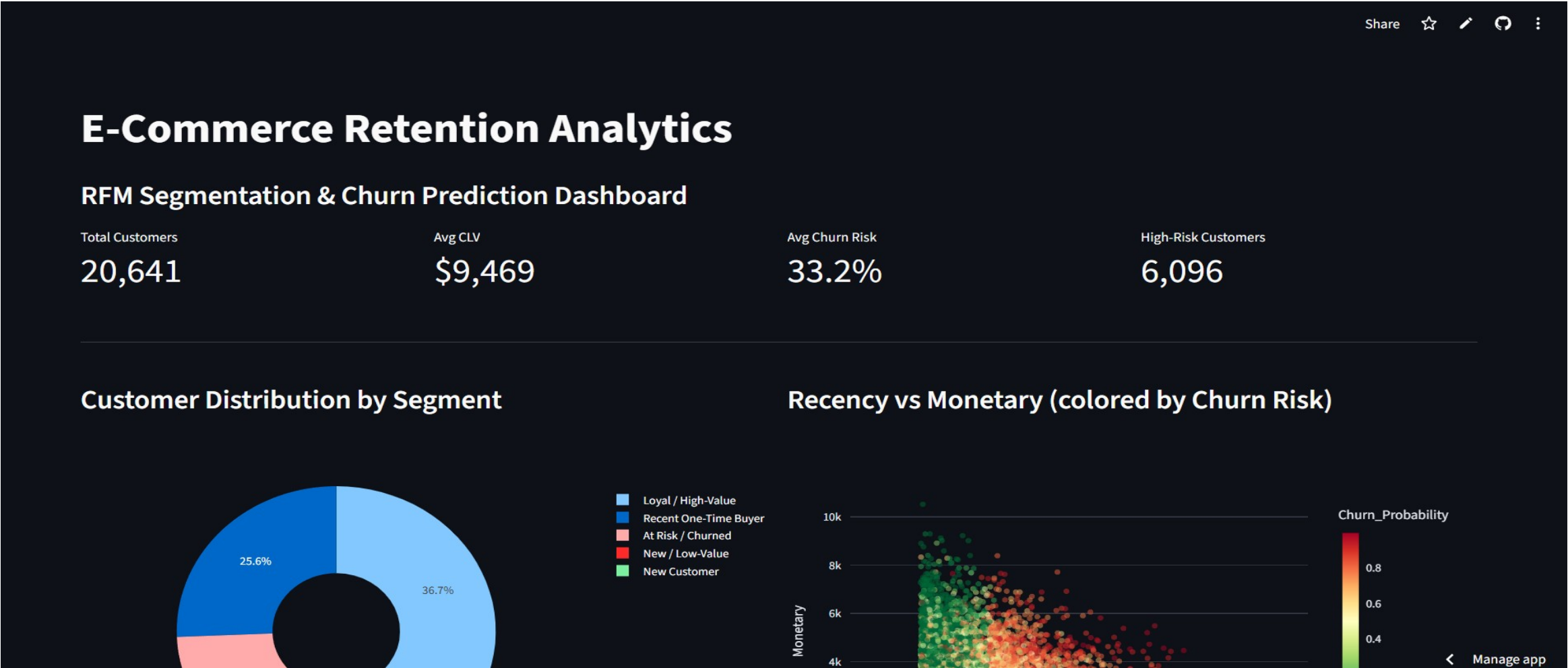
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The model's trade-off is the right one for this use case

XGBoost produces more false positives than the other models, but the cost of an unnecessary retention email is far lower than the cost of losing a customer silently.

Interactive Dashboard

Deployed on Streamlit Community Cloud



Priority Retention List

High-value, high-risk customers flagged for outreach

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 Priority Retention List (High-Value, High-Risk)

	Customer Id	Segment	Recency	Frequency	Monetary	CLV_proxy	Churn_Probability
4453	4496	Loyal / High-Value	411	6	5039.3901	30236.3403	0.9959
3160	3197	Loyal / High-Value	798	6	4571.5201	27429.1204	0.9933
10947	11039	Loyal / High-Value	574	6	4698.3801	28190.2806	0.9922
820	831	Loyal / High-Value	527	5	4519.5401	22597.7005	0.9909
3940	3980	Loyal / High-Value	704	5	4534.4501	22672.2503	0.9899
6369	6424	Loyal / High-Value	854	5	4475.2101	22376.0503	0.9885
3229	3266	At Risk / Churned	559	4	3658.6201	14634.4803	0.9856
8717	8790	Loyal / High-Value	346	7	4979.5101	34856.5706	0.9832
9535	9612	Loyal / High-Value	357	5	4451.6601	22258.3005	0.9822
3299	3337	Loyal / High-Value	364	10	5929.2901	59292.901	0.9798

2278 customers meet high-value (top 25% CLV) + high-risk (>50% churn probability) criteria

Selection criteria: top 25% CLV proxy AND churn probability above 50%. 2,278 customers meet both thresholds — this is the actionable retention target list, ranked by churn risk.

Conclusion & Next Steps

What Was Delivered

- 5-segment RFM customer model (K-Means, 20,641 customers)
- 3-model churn classifier comparison, final ROC AUC 0.907
- SQL-based cohort aggregation and CLV proxy
- Live, deployed Streamlit dashboard with a prioritized action list

Recommended Next Steps

- Pilot retention outreach on the 2,278-customer priority list
- Track campaign response to validate model-driven targeting
- Revisit churn threshold and features as more data accumulates
- Extend features with product-category diversity, discount usage

Thank you

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